Test notes

The resistance of a material in its unprotected condition to a number of standard operating environments, categorized qualitatively on the following four point scale:

Rating	Meaning
Unacceptable	Do not use in the unprotected condition
Limited use	Not recommended, although may be suitable for short term applications
Acceptable	May require additional protection
Excellent	No degradation in material performance expected after long term exposure

From CES.

Pugh Chart of Power Supply

	Solar Energy	Wind Energy	Replaceable Battery (reference)	Heat (Furnace/ combustion)	Power Outlet
Availability of the Source	-1	-1	0	1	0
Energy Production	1	1	0	0	1
Mechanical Need for Replacement	1	0	0	1	1
Portability	-1	-1	0	-1	0
Independence on Environment	1	1	0	0	0

Rating Matrix of Glass

	Yield Strength/ Compression (Mpa)	Tensile (Mpa)	Hardness (HV)	Price (USD/kg)	Fracture toughness (MPa)	Durability (UV Radiation)	Thermal expansion coefficient (µstrain/°C)	Durability (Fresh Water)
Zirconia bio-ceramic	1900 - 2000	750 - 850	1200 - 1250	18.7 - 27	9 - 10	Fair	118 - 132	Excellent
PVC (rigid, moding and extrusion)	2410 - 3300	41.4 - 52.7	12-16	1.4 - 1.6	3.63 - 3.85	Fair	90 - 180	Excellent
TPU (shore D60)	18.2 - 20.2	47.2 - 56.1	5	3.1 - 4.7	3.76 - 4.67	Fair	120 - 140	Excellent
Glass, C grade	2670 - 2750	3200 - 3300	500 - 600	4.06 - 6.81	0.5 - 1	Excellent	6.3 - 6.45	Excellent
PMMA - impact modified (Acrylic Glass)	43.9 - 96.5	34.5 - 62.1	11 - 18	2.21 - 2.44	1.81 - 2.18	Good	86.4 - 144	Excellent
PC (polycarbonate, copolymer)	76.9 - 86.9	57.2 - 69	19 - 22	3.4 - 3.64	3.83 - 4.6	Fair	70 - 137	Excellent

Pugh Chart of Glass

	Zirconia bio-ceramic	PVC (rigid, molding and extrusion)	Glass, C grade	PC (polycarbonate, copolymer)	TPU	PMMA - impact modified (Acrylic Glass) - reference
Compression	1	1	1	1	-1	0
Tensile	1	0	1	1	0	0
Hardness	1	0	1	1	-1	0
Price	-1	1	-1	-1	-1	0
Fracture toughness	1	1	-1	1	1	0
Durability (UV radiation)	-1	-1	1	-1	-1	0
Thermal expansion	0	1	-1	0	0	0
Durability (fresh water)	0	0	0	0	0	0

Red indicates that we weighted that metric more heavily.

Rating Matrix of Metals

	Yield Strength/ compression (Mpa)	Tensile (Mpa)	Hardness (HV)	Price (USD/kg)	Fracture toughness (MPa)	Durability (UV Radiation)	Thermal expansion coefficient (µstrain/°C)	Durability (Fresh Water)
Low Alloy Steel	400-1500	550-1760	140-693	0.69-0.82	14-200	Excellent	15-35	Acceptable
High Carbon Steel	400-1640	550-1160	160-650	0.64-0.77	27-92	Excellent	11-13.5	Acceptable
Cast Iron, Ductile	250-680	410-830	115-320	0.42-0.44	22-54	Excellent	10-12.5	Acceptable
Medium Carbon Steel	305-900	410-1200	120-565	0.64-0.77	12-92	Excellent	10-14	Acceptable
Stainless Steel	170-1000	480-2240	130-570	5.61-6.1	62-150	Excellent	13-20	Excellent

Pugh Chart of Metals

	Low Alloy Steel	High Carbon Steel (Ref)	Cast Iron, Ductile	Medium Carbon Steel	Stainless Steel
Compression (less important)	0	0	-1	-1	-1
Tensile	0	0	-1	-1	0
Hardness	0	0	-1	0	0
Price	0	0	1	0	-1
Fracture toughness	1	0	-1	0	1
Durability (UV radiation)	0	0	0	0	0
Thermal expansion	-1	0	0	0	-1
Durability (fresh water) --RUSTING	0	0	0	0	1

Red indicates that we weighted that metric more heavily.

Pairwise Comparison of Sensors

	Piezo (A)	Accelerometer (B)	Vibration Switch Module (C)	Distance/Range Sensors & Object Detection (D)	Tilt Switch Module (E)	Motion Sensors (F)
Piezo (A)	-	???	A	A	A	A
Accelerometer (B)	-	-	B	B	B	B
Vibration Switch Module (C)	-	-	-	C	E	C
Distance/Range Sensors & Object Detection (D)	-	-	-	-	E	F
Tilt Switch Module (E)	-	-	-	-	-	???
Motion Sensors (F)	-	-	-	-	-	-